# Niranjan Khedkar

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## Summary

AI Engineer with 3 years at IBM building and operating production GenAI systems for global enterprise clients. Hands-on across the full lifecycle: LLM fine-tuning, RAG and agentic pipelines, GPU inference optimization, containerized deployment on Kubernetes/OpenShift, and multi-cloud production operations. Delivered 15+ production AI systems generating $10M+ in revenue for Japanese and Fortune 500 clients.

## Skills

**Languages**: Python, SQL, Golang, C/C++, JavaScript, Bash

**GenAI & LLMs**: LLM Fine-tuning (LoRA, SFT), RAG, Agentic AI (CrewAI, LangGraph), LangChain, LlamaIndex, MCP, vLLM, PEFT, Prompt Engineering, Vector Databases (Milvus, ChromaDB, FAISS), LLM Evaluation (RAGAS, BLEU, ROUGE, COMET, BLEURT)

**Machine Learning**: PyTorch, TensorFlow, Scikit-learn, XGBoost, Transformers (BERT), YOLO (Object Detection), OpenCV, Multi-modal AI (Image-to-Text)

**MLOps & Cloud**: Docker, Kubernetes, OpenShift, Helm, AWS, GCP, Azure, CI/CD, Monitoring & Inference Optimization

**Data & Backend**: Pandas, NumPy, PostgreSQL, FastAPI, Flask, Django, REST APIs, Streamlit

**Spoken Languages**: English (Native), Japanese (JLPT N4, working proficiency with clients)

## Experience

**IBM** — Bengaluru, India
*AI Delivery Consultant | Applied AI* — *Nov 2025 – Present*

- Architect and operate production AI/ML workloads on OpenShift and Kubernetes, owning orchestration, autoscaling, and resource optimization for containerized inference services.
- Operate enterprise AI systems for Fortune 500 clients across logistics, enterprise software, and telecommunications, maintaining **99.9%+ uptime** for mission-critical automation workloads.
- Fine-tune deployed models and automation workflows against production feedback, improving inference speed, accuracy, and resource consumption per client SLAs.
- Troubleshoot complex production incidents across multi-cloud environments (AWS, GCP, Azure, IBM Cloud), reducing mean time to resolution via systematic debugging and runbooks.

**IBM** — Bengaluru, India
*AI Engineer | Forward Deployed Engineer (FDE)* — *Oct 2023 – Oct 2025*

- Architected and deployed **15+ production GenAI systems worth $10M+ in revenue** for Japanese enterprise clients (leading automotive manufacturers and financial institutions), using enterprise LLMs with RAG pipelines and autonomous agents.
- Fine-tuned **IBM Granite and Llama** models on domain-specific datasets (**100K+ documents**), improving response accuracy and contextual relevance by **20–30%** in production.
- Cut manual research and coding time by up to **60%** via multi-agent systems (CrewAI, LangGraph) orchestrating retrieval, summarization, and tool invocation across APIs, databases, and knowledge bases.
- Delivered AI-driven code generation and conversion pipelines for a leading global automotive manufacturer and a major consumer-electronics company, accelerating software migration and boosting developer productivity by **40%**.
- Built hybrid-cloud AI infrastructure (Docker, OpenShift) with end-to-end MLOps and rigorous LLM evaluation (perplexity, BLEU, ROUGE, human-in-the-loop) plus automated feedback loops.
- Built computer vision and multi-modal AI solutions, including image-to-text generation and YOLO-based symbol detection and classification for nuclear plant engineering diagrams.

**Richin AI** — Mumbai, India (Remote)
*AI Engineer (Freelance) | Python, NLP, Chatbots* — *Jul 2023 – Aug 2023*

- Designed and deployed a customer-facing chatbot for a finance company, cutting response time by **40%** and improving sentiment analysis accuracy by **18%**.

## Projects

**Neural Machine Translation Platform – ESA Partner** | *vLLM, NVIDIA A100, Gemma 2/3, LoRA, SFT* — May 2025 – Nov 2025

- Primary technical resource for a global language-services provider's neural translation platform; migrated the serving stack from AWS to cloud GPU infrastructure (2× NVIDIA A100 80GB) via a BYOM vLLM runtime with continuous batching.
- Designed an asynchronous proxy layer that load-balanced and dispatched batch translation requests in parallel across two vLLM model replicas, achieving **6.2× inference throughput** (2712s → 280s for 2,000 segments).
- Built an end-to-end fine-tuning pipeline with Hugging Face PEFT: corpus prep, base model benchmarking (Gemma 2 9B / Gemma 3 12B/27B), LoRA + SFT continual pre-training, and multi-metric evaluation (BLEU, COMET, BLEURT, MetricX-24 XL); delivery across three time zones **secured the ESA deal**.

**Project Hail Mary – Offline Audio Translator** | *Python, Librosa, FAISS, ONNX Runtime, Streamlit* — Personal

- Built a fully offline sound-to-word translator inspired by Andy Weir's novel: extracts 40 MFCC features per audio clip via Librosa, indexes them in FAISS, and retrieves the closest match by L2 distance.

## Education

**Indian Institute of Management (IIM) Lucknow** — Lucknow, India
*Executive Programme in AI for Business* — *Jul 2026 – Jul 2027*

**Pillai College of Engineering, University of Mumbai** — Mumbai, India
*Bachelor of Technology (CGPA: 9.61/10)* — *Jul 2019 – Jul 2023*

## Publications, Achievements & Certifications

**Certification**: **Claude Certified Architect – Foundations** (Anthropic, 2026) – Claude API, Agent SDK & MCP

**Publication** (IJSREM): *Determination of Medicinal Leaf Properties* – CNN classification (94% accuracy)

**IBM Employee Equity Program (EEP) Award** | **IBM 100% Club Member** | **JLPT N4 Certified** (Japanese)